AugSteal: Advancing Model Steal With Data Augmentation in Active Learning Frameworks

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***Abstract*— With the proliferation of machine learning models in diverse applications, the issue of model security has increasingly become a focal point. Model steal attacks can cause significant financial losses to model owners and potentially threaten the security of their application scenarios. Traditional model steal attacks are primarily directed at soft-label black boxes, but their effectiveness significantly diminishes or even fails in hard-label scenarios. To address this, for hard-label black boxes, this study proposes an active learning-based Fusion Augmentation Model Stealing Framework (AugSteal). This framework initially utilizes large-scale irrelevant public datasets for deep filtering and feature extraction to generate reliable, diverse, and representative high-quality data subsets as the stealing dataset. Subsequently, we developed an adaptive active learning selection strategy that selects data samples with significant information gain for different black-box models, enhancing the attack's specificity and effectiveness. Finally, to further address the trade-off between query budget and steal precision, this paper designed a Fusion Augmentation training method constituted of two different loss functions, enabling the substitute model to closely approximate the decision distribution of the target black box.The comprehensive experimental results indicate that, compared to the current state-of-the-art attack methods, our approach achieved a maximum performance gain of 8.21% in functional similarity for the substitute models in simulated black-box scenarios CIFAR10, SVHN, CALTECH256, and the real-world application Tencent Cloud API.**

***Index Terms*— Deep neural network security, model stealing, model extraction, data augmentation, active learning.**

I. Introduction

DEEP Neural Networks (DNNs) have garnered widespread attention due to their remarkable achievements in various fields. These achievements are underpinned by a reliance on large-scale datasets and high-performance computing resources. Consequently, a new business model has emerged: providing users with fully trained DNNs via cloud services. To preserve the intellectual property of these models, these services are typically offered through strictly defined Application Programming Interfaces (APIs). While cloud services have significantly advanced the widespread application of machine learning technologies, they also expose potential security risks.

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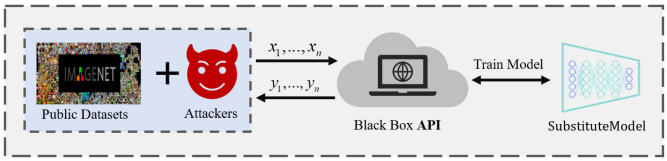


Fig. 1. Overview of model steal attacks. Attackers utilize public datasets to query the black-box model, thereby obtaining its labels. These labels are then used to train a local substitute model.

Research indicates that these APIs are vulnerable to model steal attacks, as depicted in Fig.1. Such attacks not only threaten the commercial value of the original model but also may be used to generate adversarial samples [1], [2], thereby jeopardizing the model's privacy and security. The greatest challenge in model steal research lies in the limited number of accesses to the black-box API, requiring consideration of the cost constraints of querying the API and the need to avoid triggering the model's security defense mechanisms [3], [4], [5]. Furthermore, given that online APIs can only interact through inputs and outputs, with some only returning hard-label results, this limitation increases the difficulty of executing attacks.

To overcome these limitations without increasing query costs and to train a substitute model highly similar to the target model, this paper introduces a universal model stealing framework named 'AugSteal'. Our experimental results prove that, even under extreme constraints of having no knowledge about the internal structure of the black-box model and being limited to hard-label outputs, the AugSteal attack method remains effective and demonstrates exceptional attack performance. Specifically, the main contributions are as follows:

- We proposed an innovative dataset optimization algorithm that effectively eliminates confounding factors while preserving the integrity of samples. To further enhance data quality, we introduced Class Activation Mapping (CAM) for in-depth analysis of data features, thereby generating high-quality samples with more pronounced features, providing a solid foundation for the model.
- We developed an active learning selection strategy that merges marginal probability and clustering methods (MPCL), focusing on selecting highly valuable samples for model training and ensuring broad coverage of the dataset across multiple dimensions. This strategy aims to improve the performance of the substitute model while reducing the number of labeled samples required.

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- We designed an data augmentation module named Fusion Augmentation (FusionAug), integrating two innovative loss function calculation methods and introducing a dynamic hyperparameter tuning strategy. This design allows them to work synergistically and efficiently during the model training process, thereby significantly improving the similarity of the substitute model to the target black-box model under a limited query budget.

The remainder of this paper is organized as follows: In Section II, we provide a comprehensive review of existing model steal research and related work. We clarify the problem definition and assumptions, and in Section III, we describe in detail the implementation process of our method. Through the experiments presented in Section IV, we validate the effectiveness of our approach. Finally, we conclude the paper and suggest directions for future research.

II. Related Works

A. Model Steal Attacks

In early research within the field of model steal, some methods [6], [7] required detailed knowledge of the machine learning model's architecture, limiting their extension to neural network black-box steal. Addressing this challenge, Papernot et al. [8] introduced a data augmentation approach based on Jacobian matrices, gradually approximating the decision boundaries of the target model and effectively extracting model information from a black-box setup. DRMI [9] developed an efficient optimization algorithm to obtain specific subsets of data samples. However, these methods typically needed access to datasets similar to the target model, which is often impractical in real-world attack scenarios. Consequently, Copycat CNN [10] first proposed using public datasets for neural network model steal, eliminating the need for access to the target model's original dataset. KnockoffNets [11] introduced a method for model stealing without the original training set, by exploring training data through random and adaptive strategies to find optimal data across a wide range of public datasets.

To protect the privacy and security of models, Lee et al. [12] proposed a method to perturb model predictions, preserving only the Top-1 label to resist steal attacks. Orekondy et al. [13] altered the output distribution at the expense of some model accuracy. However, these defense strategies become ineffective against attackers who only need access to hard labels. Against this backdrop, Activethief [14] broke through the limitation of relying solely on soft labels, proposing a method to steal image classification models using a single unlabeled public dataset. Building on this, Black-box Dissector [15] introduced a new method combining an erasure module and a knowledge distillation module. This approach aims to increase the information content hidden in hard labels and use the soft labels in the substitute model to eliminate overfitting.

B. Data-Free Model Stealing Attacks

To address the challenge of data scarcity, Micaelli [16] introduced a data-free model steal technique based on Generative Adversarial Networks (GANs). Following this, Zhou et al. [17] effectively mitigated the issue of uneven sample distribution in traditional GANs by improving the multi-branch GAN structure and integrating a label control module. Yoo et al. [18] demonstrated a method of reconstructing images using a teacher model, successfully extracting knowledge to guide model training in situations lacking actual data. Under limited conditions, Sanyal et al. [19] enhanced the practicality of model steal by using the gradients of a cloned network as a proxy, especially in scenarios where only hard labels are available. Truong et al. [20] emphasized the importance of choosing the appropriate loss function for accurately replicating the target model in data-free conditions.

Due to the potential significant divergence between the generated data distribution and the real training data, this leads to difficulties in achieving convergence in unstable training processes and requires substantial query budgets, limiting the practical feasibility of these data-free generation methods. To address this issue, we propose an analysis algorithm based on Grad-CAM [21]. This algorithm filters out high-quality data subsets on a wide range of public real datasets and removes noise interference, thereby significantly improving the effectiveness of model steal. In existing research, Grad-CAM has been primarily applied to enhance the transparency and interpretability of model decisions. For instance, the study [15] utilized Grad-CAM to guide the erasure of critical regions in input samples, thereby directing sample selection. These applications are largely focused on the model training phase. However, our research advances the application of Grad-CAM to the preparatory phase of model stealing, serving as a tool to systematically optimize and construct the model stealing dataset.

C. Active Learning

With the significant rise in the cost of data acquisition, active learning [22], [23] has gained substantial attention in the field of machine learning. The core of active learning strategies lies in optimally selecting samples from an unlabeled data pool that significantly enhance model performance for labeling and training, aiming to maximize model performance while maintaining minimal labeling costs. This strategy exhibits irreplaceable advantages in scenarios with limited annotation resources. Recently, [24] introduced the concept of 'Learning Loss', which guides sample selection by predicting the model's future loss, initiating a novel paradigm for sample selection. Subsequently, Ash et al. [25] proposed a deep batch active learning architecture, emphasizing the use of sample diversity and their contribution to the uncertainty in the model's gradient lower bound for data selection.

We have applied and extended the principles of active learning to the domain of model stealing, devising an active selection strategy that combines clustering with a marginal probability approach. The objective is to select samples from the stealing dataset that offer the maximum information gain. Our strategy ensures not only the diversity of samples but also considers their contribution to model uncertainty. Similarly, the DFAL strategy introduced by ActiveThief [14] also employs the principles of active learning, applying minor perturbations to each sample and inducing model misclassification

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to probe the uncertainty of samples. However, the DFAL strategy necessitates the computation of perturbations for each individual sample, which not only increases computational resource consumption but is also inefficient when processing large-scale datasets. In contrast, our MPCL strategy requires just a single forward pass of the model to rapidly calculate marginal probabilities, substantially reducing computational costs. Moreover, the MPCL strategy does not depend on specific distance metrics or feature space assumptions, facilitating easier adaptation to various target models and data distributions, and exhibiting superior computational efficiency and scalability in large-scale dataset and model applications.

D. Data Augmentation

Data augmentation, by applying a series of transformations to original data, generates new training samples, thereby mitigating model overfitting and enhancing robustness. Mixup [26] was one of the first proposed image overlay augmentation schemes, generating new training samples through linear interpolation between pairs of images, effectively expanding the distribution of training data. Subsequently, CutMix [27] combined parts of different images, with this type of regional mixture fostering the model's understanding of local patterns. However, this method could lead to the loss of complete targets, affecting the model's learning efficiency. GridMask [28] introduced an effective structured data augmentation method by applying controllable occlusion patterns to images. While this approach avoids the deletion of entire targets, the use of zero-value padding in the erased areas could lead to the loss of pixel information, which is detrimental to the model's steal of effective features.

In response to the aforementioned shortcomings, we introduce GridMix, a novel image data augmentation method. The core idea of GridMix is to selectively erase parts of an image and fill these erased areas with corresponding regions from another image. This approach not only retains the advantage of GridMask in balancing the deletion and preservation of information but also capitalizes on the benefit of CutMix, which uses real image data for filling instead of zero-value padding. This method ensures that the model is not exposed to non-informative pixels during training and can learn a richer combination of features.

Table I presents a comparative analysis between related works and our approach (AugSteal). Herein, "P" denotes problem domain data (datasets related to training with the black-box model), while "N" indicates non-problem domain data (datasets not used for training with the black-box model).

III. AugSteal

A. Problem Statement

In considering a black-box model deployed in the cloud, it can be conceptualized as a decision function f, which takes an input x and produces an output $f(x)$. To further protect the intellectual property of the model, the model owner only discloses the label associated with the highest probability in the probability vector $f(x)$. To assess the effectiveness of model stealing, we adopted the widely accepted metric of agreement accuracy. This metric evaluates the efficacy of model stealing by querying both the substitute model and the black-box model on the test set $\mathcal{D}_{\text{test}}$ of the black-box model dataset, and by comparing the agreement ratio based on hard labels. The formula for agreement accuracy is as follows:

$$\text{Agree}(f, \tilde{f}) = \frac{1}{n}\sum_{i=1}^{n}\mathbb{I}(f(x_i) = \tilde{f}(x_i)) \tag{1}$$

Here, n represents the size of the test dataset, and $\mathbb{I}$ is the indicator function, which is 1 when the outputs of the substitute model $\tilde{f}$ and the black-box model f are identical, and 0 otherwise. The effectiveness of model stealing is directly proportional to the agreement accuracy, meaning the higher the agreement accuracy, the more significant the effectiveness of the model stealing.

Algorithm 1 AugSteal Framework

Input: Stealing dataset $\mathcal{D}_{\text{steal}}$, total query budget Q, target black-box model f, querying the black box in each iteration x_i, validating dataset T, iteration limit N, label repository L_{pool}, initial data subset x_0.

Output: Substitute Model $\tilde{f}$.

Initialize $|x_i| = \frac{1}{N}(Q - |x_0| - T)$, $L_{\text{pool}}=\varnothing$, $x_0=\varnothing$, $i=1$
// **Step-Pre**: High-Quality data filtering, generate $\mathcal{D}_{\text{steal}}$
// **Step-1**: Training the initial substitution model $\tilde{f}$
Select x_0 from $\mathcal{D}_{\text{steal}}$, $\mathcal{D}_{\text{steal}} \leftarrow update(\mathcal{D}_{\text{steal}} - x_0)$
Query f for (x_0, y_0) where $y_0 = f(x_0)$ and store in L_{pool}
Utilizing the L_{pool} to train the model $\tilde{f}$ through FusionAug
// **Step-2**: Iterative training
while $i < N$ **do**
    Evaluate $\tilde{f}$ on $\mathcal{D}_{\text{steal}}$ to get predictions $(\mathcal{D}_{\text{steal}}, \tilde{f}(\mathcal{D}_{\text{steal}}))$
    Select x_i via an Active Learning Strategy, $x_i \in \mathcal{D}_{\text{steal}}$
    Update $\mathcal{D}_{\text{steal}} \leftarrow update(\mathcal{D}_{\text{steal}} - x_i)$
    Acquire true labels for x_i as $y_i = f(x_i)$
    Add labels to L_{pool}, $L_{\text{pool}} = L_{\text{pool}} + (x_i, f(x_i))$
    $\tilde{f} \leftarrow update$ (Train model $\tilde{f}$)
    $i = i + 1$
Return $\tilde{f}$

B. AugSteal Framework Overview

AuSteal is a versatile model stealing method, demonstrating outstanding steal performance in multiple black-box scenarios, with a maximum augmentation in stealing effectiveness of up to 8.21%. Utilizing active learning methodologies, we designed a three-phase model stealing process. Fig.2 illustrates the overall framework, while Alg.1 details the implementation of this framework.

Step-Pre: We conducted an in-depth study and designed an innovative dataset optimization algorithm, successfully constructing a high-quality steal dataset, denoted as $\mathcal{D}_{\text{steal}}$, from large-scale public datasets. This dataset will provide crucial support for the subsequent steal process.

Step-1: In this phase, we select an initial data subset x_0 from the steal dataset to query the black-box model f, thus obtaining corresponding labels $(x_0, f(x_0))$. These labeled data are then incorporated into the labeled data pool L_{pool}.

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TABLE I
COMPARISON BETWEEN RELATED WORKS AND AUGSTEAL (OURS)

Methods / Features	Hard labels	Black-Box	Active Learning	Data used in the Experiments	Data-Free
KnockoffNets [11]	×	✓	×	N,P	×
Copycat [10]	✓	✓	×	N,P	×
ActiveThief [14]	✓	✓	✓	N	×
Black-BoxRipper [29]	×	✓	×	N	✓
Black-box Dissector [15]	✓	✓	×	N	×
AugSteal(Ours)	✓	✓	✓	N	×

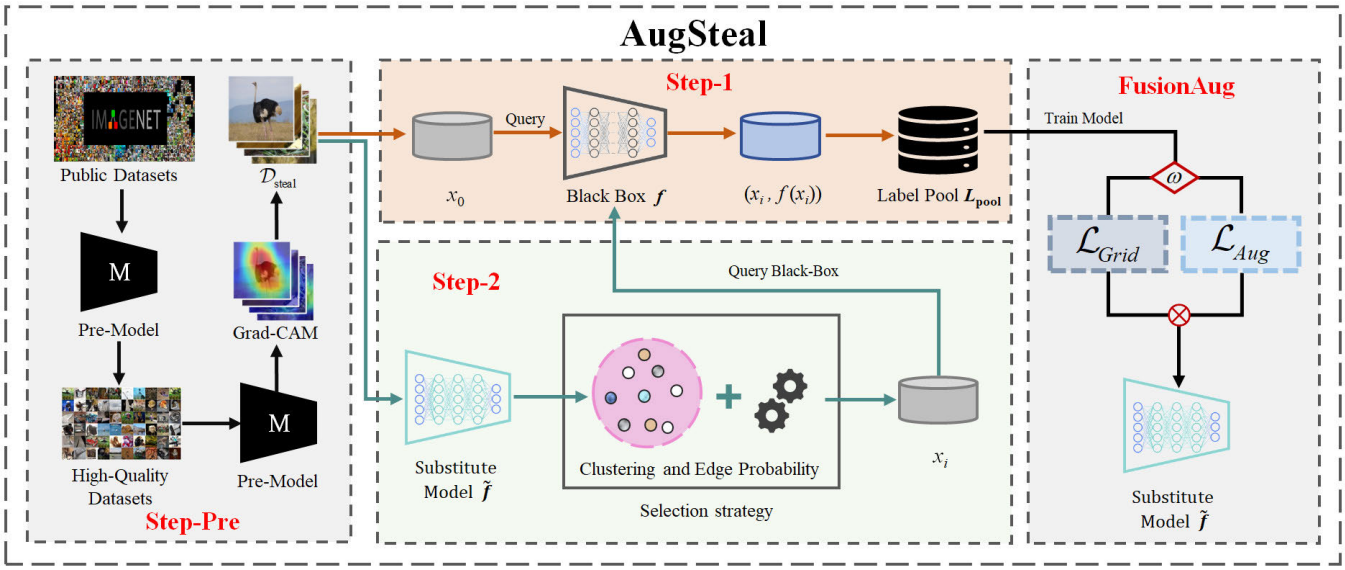


Fig. 2. AugSteal framework. Pre-Step: Building a high-quality model stealing dataset; Step-1: Creating and training an initial substitute model; Step-2: Active learning selection strategy based on margin probability and clustering; FusionAug: Training the model using a FusionAug algorithm.

Using these limited labeled data, we construct and train an initial substitute model $\tilde{f}$. To ensure diversity of x_0 and reduce redundancy, uniform sampling is performed in each high-confidence category within the steal dataset $\mathcal{D}_{\text{steal}}$.

Step-2: In this step, we use the steal dataset $\mathcal{D}_{\text{steal}}$ to query the substitute model $\tilde{f}$. Based on the predictive output of $\tilde{f}$, we propose an active learning strategy combining marginal probabilities and clustering, aimed at selecting data points x_i at the model decision boundary to optimize the substitute model. This process repeats for a fixed number of iterations N until the query budget is exhausted.

Train Model: During the model training process, we introduce an innovative fusion augmentation algorithm (FusionAug) to train the substitute model, employing two distinct loss functions to control the training process, an efficient loss computation strategy.

The specific implementations and technical details of each part will be discussed in depth in subsequent chapters. It is important to note that to prevent data duplication when querying the black-box model, each time a new x_i is selected, it is removed from the steal dataset.

C. High-Quality Data Filtering

In previous model steal research, the selection of steal datasets was often not sufficiently emphasized. Traditional methods tended to use public random datasets affected by noise, or chose core set (Coreset) samples [30] from public datasets as a starting point. Although these Coreset samples are suitable for training initial substitute models, they lack broad applicability. In our work, we observed that high-confidence samples could provide more distinct feature representations. To address these challenges, we developed a high-quality dataset optimization algorithm based on CAM to enhance the quality of training data and applied it throughout the training process. The substitute model trained with data selected by this method showed a 2.71% increase in accuracy compared to models trained with the original dataset. Fig.2 outlines the preprocessing steps (Pre-Step) of the algorithm, while Fig.3 provides a visual representation of the algorithm's steps.

In the design process of the high-quality dataset filtering algorithm, we initially input the public dataset $\mathcal{D}_{\text{public}}$ into the deep learning model M, obtaining the predictive probability for each category. The model M processes each image $\mathcal{D}^m_{\text{public}}$ through a series of operations such as convolution, pooling, activation functions, etc., ultimately generating an original output vector $Z = [z_1, z_2, \ldots, z_C]$, where C represents the number of categories.

Subsequently, we use the Softmax function to transform this original output vector Z into a probability output $P = [p_1, p_2, \ldots, p_C]$. In this transformation, each element p_c represents the probability that the model predicts the image $\mathcal{D}^m_{\text{public}}$ belongs to category c. This process is expressed by

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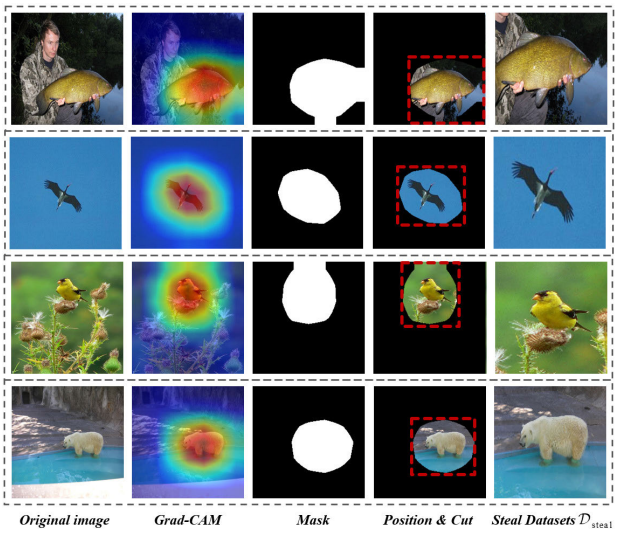


Fig. 3. High-quality data filtering.Based on high-confidence images, the initial step involves identifying sensitive regions through color depth differences, leading to the generation of masks that are superimposed onto the original images. Following this, irregular bounding rectangular areas are extracted from these masked images. Finally, these extracted regions are resized to a uniform dimension, resulting in the creation of the steal dataset $\mathcal{D}_{\text{steal}}$.

the following formula:

$$p_c = \frac{e^{z_c}}{\sum_{j=1}^{C} e^{z_j}} \quad \text{for} \quad c = 1, 2, \ldots, C \tag{2}$$

This formula means that for each category c, the probability p_c of the model M predicting the image $\mathcal{D}^m_{\text{public}}$ belongs to that category is equal to the ratio of the exponential form of the original output for category c, z_c, to the sum of the exponential forms of the original outputs for all categories. This process ensures that the sum of the probabilities for all categories equals 1.

We set a threshold τ and select those data whose model output prediction probability exceeds this threshold as high-confidence samples. These selected high-confidence samples are integrated into subset D. Specifically, to ensure the diversity and sparsity of the resulting dataset, we uniformly selected a fixed number of data at each high-confidence level, thereby constructing a simplified subset. The specific selection criteria are as follows:

$$D_m = \begin{cases} 1 & \text{if } \max\limits_{c=1}^{C} P_c(\mathcal{D}^m_{\text{public}}) > \tau, \\ 0 & \text{otherwise.} \end{cases} \qquad m \in 0, \ldots, L \tag{3}$$

where L represents the length of the large-scale public dataset.

To refine and obtain a higher-quality steal dataset, we adopted the gradient-based CAM method for deep feature visualization of subset D. This method enables us to identify and analyze sensitive and prominent features in the samples. By combining mask masking technology, we successfully located key feature areas within the images, thereby filtering out and constructing our optimized steal dataset $\mathcal{D}_{\text{steal}}$.

In particular, the stealing dataset $\mathcal{D}_{\text{steal}}$ exhibits universality, not being confined to any specific target black-box model. This is attributable to the hierarchical feature extraction capability inherent in deep learning models. Models achieve effective prediction across different data distributions through internal high-level feature representations, which are often shared via similar activation patterns among various layers of the network. The stealing dataset we generated provides a broad range of samples, constituting a rich feature space. Despite some discrepancies, the generalization ability of deep learning models and their reliance on high-level feature representations ensure the effectiveness of our constructed stealing dataset.

D. Active Learning Selection Strategy-MPCL

To mitigate the significant costs associated with direct queries to the black-box model, we incorporated an active learning strategy into the model stealing process. While existing active learning strategies such as Entropy and k-center have been extensively studied, their effectiveness in practical applications is inconsistent, sometimes even inferior to random selection. To improve this situation, this paper proposes an active learning selection strategy that fuses marginal probabilities with clustering methods, achieving up to a 2.28 percentage point increase. Specifically, in each round of iterative training, we select a set of data points and query the black-box model f to obtain their true labels. The acquired label data are then added to the labeled data pool to support the training of the substitute model $\tilde{f}$. Detailed steps are presented in Fig.4.

MPCL initially uses the substitute model $\tilde{f}$ for a prior query of the steal dataset $\mathcal{D}_{\text{steal}}$. For each data sample's output, we obtain not only its predictive probability but also further acquire hard label decisions. Based on these decision outputs, we employ a clustering algorithm to divide $\mathcal{D}_{\text{steal}}$ into multiple subsets, each representing a unique output category.

Specifically, for a data point $d_i \in D_{\text{steal}}$ queried against $\tilde{f}$, we obtain its predictive probability vector $P(d_i) = \tilde{f}(d_i)$, satisfying:

$$\sum_{c} P_c(d_i) = 1 \tag{4}$$

Here, $P_c(d_i)$ represents the probability of d_i being predicted as category c. We identify the category label corresponding to the highest probability in each data point's predictive vector:

$$y_i = \arg\max_{c} P_c(d_i) \tag{5}$$

This represents the most likely category for d_i as predicted by the substitute model $\tilde{f}$. Based on this, data points with the same hard label are grouped into the same subset:

$$D^c_{\text{cluster}} = \{d_i \in D_{\text{cluster}} | y_i = c\} \tag{6}$$

Here, D^c_{cluster} is a subset containing all data predicted as category c by the substitute model $\tilde{f}$. In this manner, we effectively divide the original steal dataset into several subsets.

When selecting data samples for active learning, a category-based selection strategy can only satisfy the requirement for sample diversity. To more precisely identify data points that contribute more significantly to model training, it is essential to explore the uncertainty in the substitute model's predictions for each data point. Here, we introduce for the first time the concept of marginal probability $P_{\text{margin}}(d_i)$, which calculates the difference between the probability of the most likely

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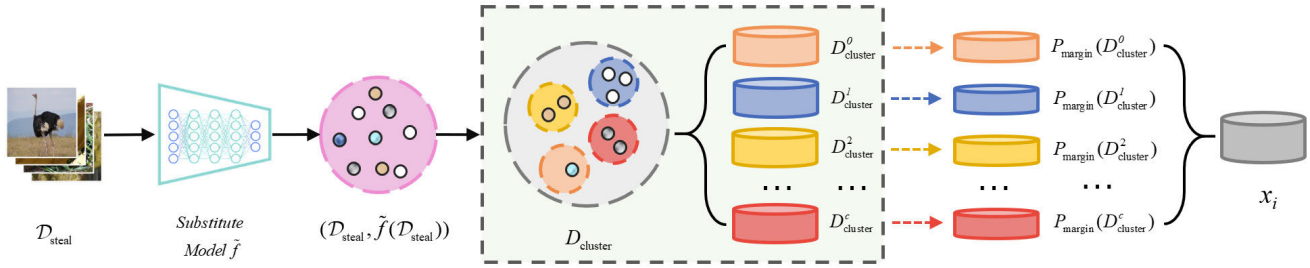


Fig. 4. MPCL.The steal dataset $\mathcal{D}_{\text{steal}}$ is used to query the substitute model, followed by data clustering based on the model's hard label outputs. Using the edge probability formula (Eq.7), uncertain data x_i within each category are identified for querying the black-box model to obtain the true labels.

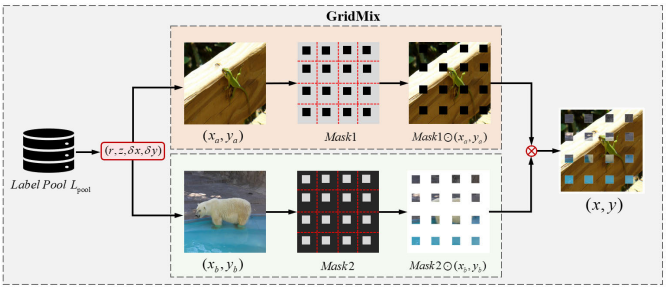


Fig. 5. This image shows examples of GridMix. Two complementary and distinct masks, formed using variables $(r, z, \delta x, \delta y)$, are employed. These masks are individually multiplied with x_a and x_b, and the resulting products are subsequently summed together.

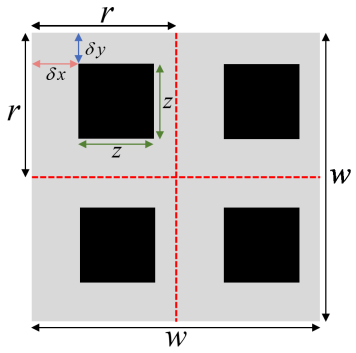


Fig. 6. Mask. The original image is divided into a grid of $\Phi \times \Phi$ squares. Within each square, a unique mask is generated based on the mask size defined by the parameter z and random offsets δx and δy. Areas of different colors in the mask are filled with segments from two distinct images, resulting in the formation of a composite image.

category predicted by the model for data point d_i and the average probability of all other categories, expressed as:

$$P_{\text{margin}}(d_i) = \max_c P_c(d_i) - \frac{1}{k-1}\sum_{c=1}^{k} P_c(d_i) \tag{7}$$

where $P_c(d_i)$ represents the probability of sample data belonging to category c, and k represents the number of categories. $\max_c P_c(d_i)$ indicates the probability of sample d_i belonging to the most likely category. Therefore, a larger marginal probability indicates greater confidence in the model's prediction for that data point, while a smaller value suggests greater uncertainty in the prediction.

E. FusionAug

To construct a more consistent model within a limited query budget, this paper introduces the FusionAug data augmentation module. Its core concept is to maximize the utilization of existing labeled data while avoiding additional queries to the black-box model. This approach not only effectively addresses the potential overfitting problem inherent in traditional augmentation techniques but also provides an efficient solution for training better-performing models under resource-constrained conditions. FusionAug consists of two different loss calculation methods: GridMix and MulAug. Under the same query budget constraints, it significantly enhances the model's performance and robustness, with performance gains of up to 5.47%.

*GridMix:*This is a novel mask-based image synthesis technique, leveraging the exchange of information blocks to create new augmented images, enhancing the model's ability to capture details and improving the robustness of training. Furthermore, by partially cropping and reorganizing real data, rather than simply filling with zero values, this technique avoids introducing non-informative pixels during training. For the construction of composite images, two images are randomly selected from a labeled data pool L_{pool}, designated as (x_a, y_a) and (x_b, y_b) respectively. Subsequently, two complementary masks, $Mask1$ and $Mask2$, are generated. By applying these masks to the original images through element-wise multiplication, a new blended image (x, y) is formed. The visualization of this synthetic process is detailed in Fig.5.

The design of the mask adopts a grid-like structure, producing mask blocks within equally divided grids. We use four parameters $(r, z, \delta x, \delta y)$ to form a unique mask, with each mask formed by tiling units, as shown in Fig.6. The specific parameters of the mask are defined as follows:

- w: The side length of the input image. In our study, we use images of size 32×32 pixels as an example.
- r: Determines the size of the grid blocks the input image is divided into, satisfying $r \times \Phi = w$, where $\Phi \geq 1$.
- z: Determines the actual size of the mask in each grid block, with a value range of $0 \leq z \leq r$.
- δx, δy: Represent the random offsets of the mask in the horizontal and vertical directions, respectively, ensuring that the mask can exhibit diverse positional transformations on the image plane, satisfying $0 \leq \delta y, \delta x \leq (r - z)$.

After determining the above parameters, we first generate a mask $Mask1$. In this mask, regions with a grayscale value

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of 1 indicate that the pixels of x_a will be retained, while those with a value of 0 indicate that the pixels of x_b will be retained. For ease of operation, we generate a mask $Mask2 = 1 - Mask1$ with opposite grayscale values to $Mask1$. Then, we use $Mask1$ to perform Hadamard product (element-wise multiplication) with x_a, and similarly with $Mask2$ and x_b. Finally, by combining these two processed images, we form a new synthetic image x.

$$x = Mask1 \odot (x_a) + Mask2 \odot (x_b) \tag{8}$$

where $\odot$ denotes element-wise multiplication.

During the synthesis process, the ratio of the mask size z relative to the grid size r directly affects the representation weight $\rho = r/z$ of each source image in the final image. Specifically, the pixel retention ratio of source images x_a and x_b is directly proportional to the square ratio of the mask size z to the grid size r. This weight distribution allows us to balance the features of different images in the synthetic image, thereby providing richer feature representations when training the model. This leads to our first loss calculation method $\mathcal{L}_{\text{Grid}}$:

$$\mathcal{L}_{\text{Grid}} = \mathcal{L}_{CE}(\tilde{f}(x), y_a)(\rho)^2 + \mathcal{L}_{CE}(\tilde{f}(x), y_b)(1 - (\rho)^2) \tag{9}$$

$\tilde{f}(x)$ represents the full probability output of the substitute model for the synthetic sample x, and $\mathcal{L}_{CE}$ is the cross-entropy loss.

MulAug: It enriches the dataset by applying a series of random transformations to each batch of input images x, including but not limited to translation, rotation, shearing, and color adjustment. The goal is to generate a collection of samples that are visually distinct, yet maintain the same semantic label y. Through this method, each batch of images is endowed with different visual representations, greatly enriching the diversity of the dataset and the generalization foundation of the model.

The core advantage of MulAug lies in its diversified transformation strategy. Each data augmentation transformation introduces random parameters, ensuring that every transformation can generate uncertainty within a predefined range. This randomness within the defined range is applicable to transformations such as rotation, translation, shearing, scaling, and color adjustment, guaranteeing the diversity of the augmented samples. MulAug fine-tunes the randomization process to ensure that each generated sample contributes a unique learning signal to the model training, with each transformation being independently random. To quantify the training effect of these augmented samples, MulAug employs the cross-entropy loss function to compute the loss, encouraging the model to better capture and optimize classification boundaries during the learning process. The specific computation method for the loss $\mathcal{L}_{\text{Mul}}$ is as follows:

$$\mathcal{L}_{\text{Mul}} = \mathcal{L}_{CE}\left(\tilde{f}\left(\sum_{i=1}^{\mu} \text{RandomAug}(x, i)\right), \sum_{i=1}^{\mu} y\right) \tag{10}$$

where μ is the number of random data augmentations.

To overcome the biases and potential adverse effects on the substitute model training associated with relying on a single loss function, this study adopts a dynamic loss selection strategy. This strategy introduces a random variable γ and sets a threshold ω, allowing the model to switch between two loss functions based on predefined criteria before training. The dynamic loss function $\mathcal{L}_{\text{AugSteal}}$ can be expressed as:

$$\mathcal{L}_{\text{AugSteal}} = \begin{cases} \mathcal{L}_{\text{Grid}}, & \text{if } \gamma = \text{random}(0, 1) \geq \omega \\ \mathcal{L}_{\text{Mul}}, & \text{other wise} \end{cases} \tag{11}$$

Specifically, during the initialization phase of model training, we first set a threshold ω for the loss function. Subsequently, in each training epoch, we generate a random variable γ that follows a statistical distribution and is constrained within the closed interval [0,1]. This variable is then compared with the predefined threshold ω to determine the application method of the loss function. If the value of γ is not less than the threshold ω, the loss function $\mathcal{L}_{\text{Grid}}$ will be used during the training process; if the value of γ is less than the threshold ω, the loss function $\mathcal{L}_{\text{Mul}}$ will be applied. In subsequent research, we will delve into how the setting of the threshold ω affects the sensitivity of model training to steal attacks.

IV. Experiments

A. Experimental Setting

Steal Dataset: This paper initially employed a high-quality data filtering algorithm based on pre-trained models and CAM on the publicly available ILSVRC2012 dataset, which includes 1,000 categories with a total of 1.2 million images. This process resulted in the generation of a data subset comprising 121,186 images, also encompassing 1,000 categories. We define this subset as the lsteal dataset, denoted as $\mathcal{D}_{\text{steal}}$.

Black-Box Model: The black-box models used in this study are trained on the CIFAR10, SVHN, and CALTECH256 datasets. These datasets are widely used in the field of image classification research and have been verified to present considerable challenges. The novelty of this study lies in selecting the most challenging hard-label black-box models for model stealing tasks to date, aiming to validate the effectiveness and applicability of our method in API environments.

Model Architectures: For comparison purposes, this paper employs the same architecture (ResNet-34) as used in [15], which achieves deeper feature steal and model optimization through residual connections and stacked convolutional layers. In the subsequent experimental section, we will explore the specific impact of different model architectures on the experimental results.

Training Regime: We use an SGD optimizer with default hyperparameters and a cosine learning rate scheduler, starting from 0.1. All training processes are iterative, and for each dataset, experiments were conducted under total query budgets of 5K, 10K, 15K, 20K, 25K, and 30K, with 20% reserved for validation and 15% as initial seed samples.

Pre-Trained Model M: In this study, our pre-trained model M utilizes the ResNet-50 model, which has been pre-trained on the ImageNet dataset. The ResNet-50 model achieves a Top-1 accuracy of 75.36% and a Top-5 accuracy of 92.13% on the dataset's validation subset. Furthermore, the pre-trained weights of the model are readily accessible from the official model repository.

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TABLE II
AGREEMENT ACCURACY (%) OF SUBSTITUTE MODELS UNDER DIFFERENT TARGET BLACK BOXES(30K)

Black-Box	CIFAR10		SVHN		CALTECH256	
	Agreement(%)	Acc(%)	Agreement(%)	Acc(%)	Agreement(%)	Acc(%)
KnockoffNets [11]	73.81	73.44	84.57	84.05	55.63	54.85
Copycat [10]	73.27	72.50	83.79	83.02	54.36	53.81
ActThief(Entropy) [14]	74.22	73.05	89.99	89.18	54.28	52.35
ActThief(K-center)	73.50	72.22	80.23	78.84	58.23	56.62
Black-BoxRipper [29]	73.65	72.28	81.59	80.18	60.57	58.45
Dissector(K-center) [15]	79.36	78.83	90.09	89.37	63.76	62.71
Dissector(Random) [15]	80.12	78.92	92.68	91.39	59.47	58.39
AugSteal(Ours)	**82.29**	**80.76**	**94.19**	**92.15**	**65.27**	**63.74**

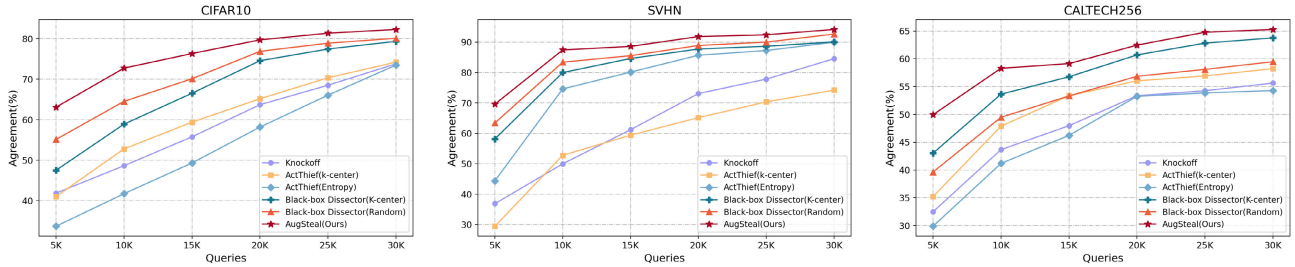


Fig. 7. Agreement accuracy curves of substitute models under different target black boxes. The X-axis represents the query budget and the Y-axis represents the agreement accuracy.

High Confidence Threshold τ: A high-confidence threshold τ is utilized to select high-quality samples, laying the groundwork for subsequent feature extraction steps. Specifically, in this experiment, τ is set to 0.95, meaning that a sample is only included for consideration if the maximum prediction probability value by the pretrained model M reaches or exceeds 95%. The determination of threshold τ is based on a preliminary analysis of the probability distribution output by the pretrained model and a reasonable comparison with baseline papers. We discovered that setting τ to 0.95 enables the selection of a subset from the publicly available dataset ILSVRC2012 that is consistent in size with the subsets used in baseline papers. This subset not only satisfies the broad representativeness of categories but also complies with the principle of data balance.

B. Black-Box Attack Results

To comprehensively evaluate the efficacy of the proposed method, this study compares it against contemporary model steal attack techniques under a query budget of 30K. The experimental outcomes are summarized in Table II, where "Agreement" refers to the degree of agreement in predictions between the target and the attacked black-box models, and "ACC" denotes the performance of the generated substitute model on the test set. Overall, the proposed method significantly outperforms comparative methods in both consistency and test accuracy. To ensure the reliability and robustness of the experimental results, each experiment was independently conducted three times, with the average calculated to determine the final outcome.

For an in-depth analysis of the improvement in substitute model consistency under different query budgets, this study meticulously compares some of the advanced methods [11], [14], [15] in the domain of model steal. Specifically, experiments were carried out on three distinct datasets, with varying query budgets set at 5k, 10k, 15k, 20k, 25k, and 30k. The results are exhaustively collated and displayed in Fig.7. A holistic analysis from the figure reveals that the proposed AugSteal method demonstrates exceptional performance across multiple black-box models, significantly outshining comparative methods in terms of consistency accuracy. Remarkably, AugSteal achieves a substantial boost in consistency accuracy while maintaining the same query cost. For instance, on the CIFAR10 dataset, even with only 8% (10K) of the total data volume as the budget, AugSteal's consistency accuracy surpasses the existing best method, Dissector, by 8.21%, further validating our method's significant advantage under constrained query budgets.

Compared to gradient-matching-based methods, AugSteal takes into account both data distribution and model decision boundaries, enabling more effective capture of the black-box model's decision patterns under limited queries, thus enhancing the efficiency and outcome of the steal attack. In summary, the experimental results validate that the AugSteal method can significantly improve the agreement accuracy of the substitute model without increasing the query budget for the black-box model.

C. Ablation Study

The ablation studies conducted in this research were aimed at quantitatively assessing the independent contributions of each algorithmic component to the model's performance and exploring their relative importance in overall performance enhancement. Notably, in the comparative analysis of the

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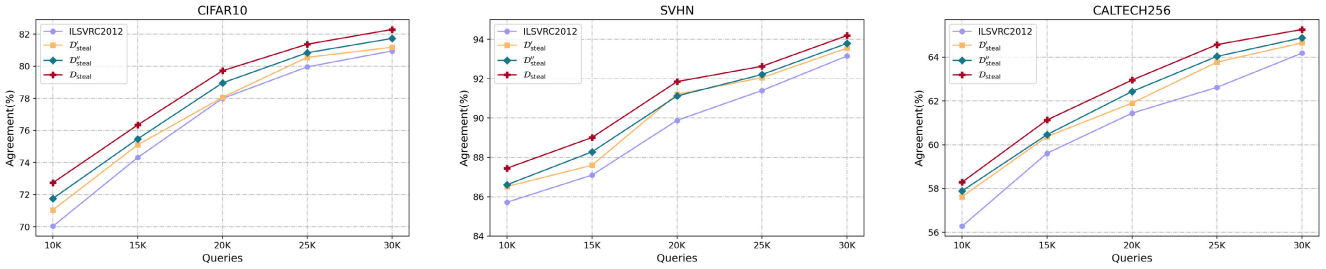


Fig. 8. The impact of different stealing datasets appropriated for model steal, including $\mathcal{D}'_{\text{steal}}$, $\mathcal{D}''_{\text{steal}}$, $\mathcal{D}_{\text{steal}}$ and ILSVRC2012.

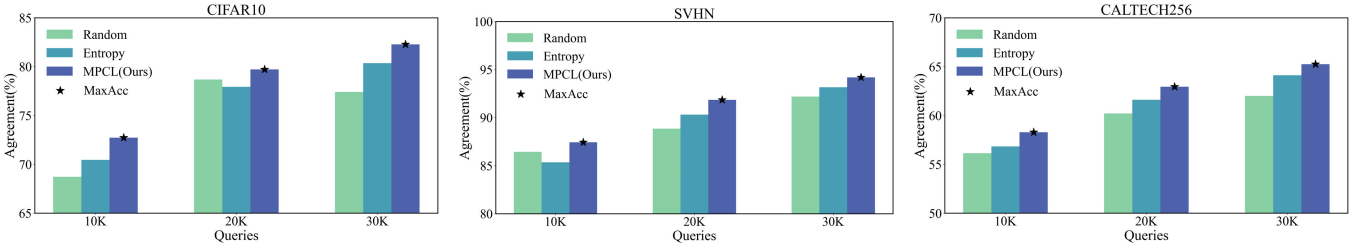


Fig. 9. The impact of active learning selection strategy-MPCL, and optimal steal result for the MaxAcc marker.

ablation study, we employed a rigorous controlled variable method: in the entire model steal process, each experiment involved removing only a single component being evaluated at a time, to measure its specific impact on overall performance.

Impact of $\mathcal{D}_{steal}$: In this study, we conducted several experimental validations using our specially constructed steal dataset $\mathcal{D}_{\text{steal}}$ on three different datasets: CIFAR10, SVHN, and CALTECH256, under various query budget scenarios. As illustrated in Fig.8, $\mathcal{D}_{\text{steal}}$ demonstrated significant stability and efficiency across all budget conditions. Particularly noteworthy is the performance on the CIFAR10 dataset, where steal operations utilizing our dataset, under a mere 10k query budget, outperformed those using the publicly available ILSVRC2012 dataset by 2.71 percentage points. The application of this approach in model steal clearly proves the pivotal role of meticulously designed datasets in enhancing steal efficiency and effectiveness. Our experimental results offer valuable references and insights for future research in this field.

The Impact of High-Confidence Samples: In the process of constructing the data set $\mathcal{D}_{\text{steal}}$ for model stealing, this study meticulously selected high-confidence samples from the ILSVRC2012 data set as a foundation. To delve into the dependency on high-confidence samples during the CAM feature extraction stage, we constructed a derivative data set, $\mathcal{D}'_{\text{steal}}$, based on randomly chosen samples from ILSVRC2012 for feature extraction. As illustrated in Fig.8, experimental results indicate that the performance of $\mathcal{D}_{\text{steal}}$ built on high-confidence samples generally decreases under various query budget constraints. Taking a 10k query budget on the SVHN as an example, $\mathcal{D}'_{\text{steal}}$ demonstrated a reduction of 0.93 percentage points compared to $\mathcal{D}_{\text{steal}}$, yet its performance still exceeded the ILSVRC2012 baseline by 0.80 percentage points. These findings underscore the significant value of high-quality data in model training and its potential advantages in optimizing model stealing attacks.

The Impact of CAM Feature Extraction: To investigate the impact of Grad-CAM's accuracy in localizing key feature areas on the model stealing process, we introduced a localization error by adding random vertical shifts to the key feature areas determined by Grad-CAM, simulating scenarios where key area localization is inaccurate. The shift distance is randomly generated based on a pre-set error range to ensure the generalizability of the experimental results. To prevent the shifted areas from exceeding the original image boundaries, we designed a boundary checking mechanism that calculates whether an area will be out of bounds before shifting; if so, it is adjusted to the closest boundary position. In Fig.10, we present two different data samples processed in this manner. Consequently, a new stealing dataset $\mathcal{D}''_{\text{steal}}$ was generated, and the experimental results were summarized along with Fig.8. We observed that compared to $\mathcal{D}_{\text{steal}}$, there is a decrease in the model's agreement accuracy when the CAM localization is offset, emphasizing the critical nature of the feature extraction step. Further analysis showed that the performance decline of $\mathcal{D}'_{\text{steal}}$ exceeded that of $\mathcal{D}''_{\text{steal}}$, further verifying the necessity of relying on high-confidence data. This indicates that even if there are errors in the feature extraction step, the quality of the underlying data remains key.

Impact of MPCL: In the realm of active learning research, we innovatively proposed the MPCL sample selection strategy. This strategy focuses on enhancing the diversity and uncertainty of samples queried from the black-box model, significantly optimizing query efficiency. As illustrated in Fig.9, we conducted detailed experimental evaluations across multiple datasets under various query budget conditions. The results demonstrated that MPCL excelled in improving model accuracy, with performance gains reaching up to 2.28% and a minimum improvement of 1.13%, compared to traditional random sampling and entropy-based selection methods. Notably, despite entropy-based methods not always performing optimally, sometimes even less effectively than random selection,

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TABLE III
THE IMPACT OF FUSIONAUG ON STEALING RESULTS (%)

Black-Box	Method	Number of Queries				
		10K	15K	20K	25K	30K
CIFAR10	No-FusionAug	67.27	71.58	75.51	78.33	80.47
	FusionAug(Ours)	**72.74**	**76.34**	**79.72**	**81.37**	**82.29**
SVHN	No-FusionAug	84.02	85.72	89.42	90.35	92.81
	FusionAug(Ours)	**87.45**	**88.61**	**91.85**	**92.43**	**94.19**
CALTECH256	No-FusionAug	54.78	56.30	60.13	63.34	64.11
	FusionAug(Ours)	**58.29**	**59.13**	**62.45**	**64.77**	**65.27**

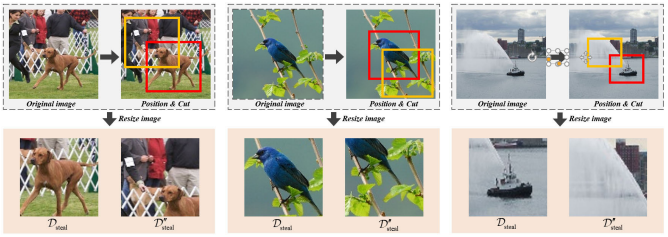


Fig. 10. During the CAM feature extraction process, a localization error was introduced, where the red box indicates the accurate feature localization, and the yellow box represents the feature localization after the shift has occurred.

our MPCL strategy consistently exhibited the best performance under all conditions, underscoring its significant superiority in active learning selection strategies. The core advantage of MPCL lies in its ability to precisely identify and select key samples near the model's decision boundary, providing highly targeted data support for model training. This approach not only markedly enhanced the efficiency of model steal but also played a crucial role in augmenting the generalizability of the substitute model, fully demonstrating the exceptional performance and practical value of our selection strategy.

Impact of FusionAug: In this segment, we conducted a comprehensive evaluation of the FusionAug algorithm across various datasets and a range of query budgets. Detailed comparisons between FusionAug and traditional training methods are elaborated in Table III. Our findings reveal that, under all black-box dataset conditions, FusionAug consistently outperformed standard supervised learning approaches. For instance, on the CIFAR10 dataset under a constrained 10k query budget, the application of FusionAug elevated the model accuracy from 67.27% to 72.74%, achieving a significant agreement enhancement of 5.47 percentage points. Similarly, under other query budget settings such as 15k, 20k, 25k, and 30k, FusionAug demonstrated agreement improvements of 4.76%, 4.21%, 3.24%, and 2.16%, respectively. These results not only underscore the marked effect of FusionAug in boosting the agreement accuracy of substitute models but also highlight its pivotal role in enhancing the training process of model steal. Furthermore, the application of FusionAug is not confined to the realm of model steal; it holds potential in a broader context of supervised learning training, offering valuable new insights and references for the field of data augmentation.

Impact of GridMix: This study examines the performance differences of the AugSteal model when employing the GridMix strategy in comparison with CutMix and GridMask. To reduce experimental uncertainty, we did not introduce the $\mathcal{L}_{\text{Mul}}$ loss term from FusionAug during training to enhance the stability of the results. The experiment compared the effects of three data augmentation techniques: GridMix, CutMix, and GridMask on the substitute models, as detailed in Fig.11. The results show that GridMix consistently outperforms the others in all experiments. This performance is primarily attributed to the improved feature density and quality of $\mathcal{D}_{\text{steal}}$ after CAM processing. In contrast, CutMix may confuse feature information due to the mixing of large sample areas, while the zero-value padding of GridMask leads to further information loss, which to some extent weakens the representational ability of the stealing dataset.

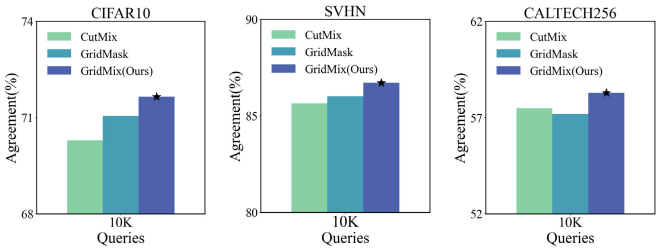


Fig. 11. Impact of GridMix, CutMix, and GridMask.

TABLE IV
EXPLORING THE PROPORTION OF THE INITIAL SEED SAMPLES

Queries	5%	10%	15%	20%	25%	30%
20K	74.83	75.14	**75.51**	75.20	74.95	74.71
30K	79.88	80.17	**80.47**	80.04	79.73	79.55

Exploring the Proportion of the Initial Seed Samples: In this section, we explore the impact of the selection proportion of the initial data subset x_0 on the active learning framework in the domain of model stealing. To control other uncertain factors, this experiment was conducted on the CIFAR10 dataset with 20k and 30k samples, using only the $\mathcal{D}_{\text{steal}}$ dataset and the MPCL selection strategy. The experimental results Table IV indicate that the best performance is achieved when the selection proportion is around 15%. We did not validate proportions below 5% or above 30%, primarily based on the following hypotheses: When the proportion is too small, the initial surrogate model's capacity to recognize is too low, leading to a slow start for the active learning framework, possibly requiring a large number of iterations to take effect; when the proportion exceeds 30%, the relevance of applying the active learning framework diminishes, as it cannot dynamically select more samples to query the black-box model based on the surrogate model's output, leaving too few options for active learning to make significant improvements.

Impact of Substitute Model Structure: This section comprehensively evaluates the impact of different substitute model architectures on fidelity accuracy, with detailed results presented in Table V. We selected the ResNet-34 structure, which is the same as the black-box model, as our baseline model and subsequently compared the performance of ResNet-18, VGG-16, ResNet-50, ResNet-101, and WideResNet-28

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TABLE V
IMPACT OF SUBSTITUTE MODEL STRUCTURE(%)

Black-Box	Queries	Substitute Model Structure					
		ResNet-34	ResNet-18	ResNet-50	WideResNet-28	ResNet101	VGG-16
CIFAR10	10K	72.74	71.49	72.63	**73.53**	72.08	64.77
	20K	79.72	78.05	79.08	**80.28**	78.43	71.54
	30K	82.29	80.47	82.13	**83.41**	82.01	73.95
SVHN	10K	87.45	85.13	88.21	**88.89**	86.75	78.37
	20K	**91.85**	90.72	91.56	91.78	90.87	82.17
	30K	94.19	92.58	94.33	**94.41**	94.12	86.78
CALTECH256	10K	58.29	57.01	58.84	**59.38**	57.59	51.15
	20K	62.45	60.24	62.47	**62.89**	62.07	54.51
	30K	65.27	64.55	**66.76**	65.85	66.28	57.49

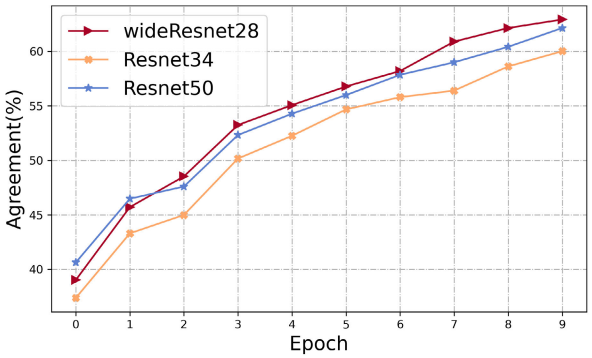


Fig. 12. AugSteal conducted model stealing on real-world APIs and validated the impact of different substitute model structures on the results.

(width factor of 2). The findings indicate a clear advantage in fidelity accuracy when the architecture of the substitute model is similar to that of the black-box model. For instance, across various datasets, the ResNet architecture outperformed VGG. Moreover, the depth and complexity of the networks play a pivotal role in enhancing fidelity accuracy. Taking the CIFAR10 dataset as an example, ResNet-50 achieved a fidelity accuracy of 72.63% with a query budget of 10K, which is 1.14 percentage points higher than that of ResNet-18. However, further experiments demonstrate that continuously increasing the depth of the model architecture does not necessarily lead to sustained improvements in fidelity accuracy. In fact, more complex model structures can sometimes result in decreased fidelity due to overfitting. For example, on the SVHN dataset with a query budget of 10K, the fidelity accuracy of ResNet-101 was 1.46 percentage points lower than that of ResNet-50. When the query budget was increased to 30K, although the additional number of labels mitigated the effects of overfitting, the fidelity accuracy of ResNet-101 still declined.

D. Stealing Functionality of a Real-World API

In the realm of model stealing attacks, the challenges are especially pronounced when confronting real-world APIs, since attackers are typically unable to acquire precise information about the target model. This research affirms the efficacy of the AugSteal method with respect to real application APIs. We employed Tencent's TI-ONE[1] machine learning platform for our experiments, which provides users with integrated services from data input and model training to management and deployment, using a publicly available waste classification dataset.[2] During the model stealing process, we only had access to the model's prediction results (top-1 output), without any internal details of the target model. The study utilized the $\mathcal{D}_{\text{steal}}$ dataset as the attack set, with 20% of the data allocated as the test set under a 30K query budget. The detailed experimental results, showcased in Fig.12, include a highest achieved agreement accuracy of 63.71% on the test set, remarkably attained with a limited cost of approximately 13 USD. These findings not only confirm the efficiency of the AugSteal method but also highlight its wide applicability in practical scenarios. It also serves as a caution to machine learning service providers about the risks of complex models and powerful APIs being maliciously attacked and replicated.

E. Investigation of the Optimal Threshold Value ω

This study evaluated the impact of dynamically adjusting loss function weights on model performance. The experiments employed two types of loss functions, $\mathcal{L}_{\text{Grid}}$ and $\mathcal{L}_{\text{Mul}}$, by altering the threshold parameter ω to change their weight ratio during training. The research was conducted on a Cifar10 black-box model under training instances of 10k, 20k, and 30k, with results summarized in Table VI; only the ω value was adjusted during the experiments. The experiments demonstrated that with a small number of labels (such as 10k), in order to train with better consistency, the loss function tends to choose a larger ω value (e.g., $\omega = 0.7$), at which point the model is more inclined towards $\mathcal{L}_{\text{Mul}}$ because its random augmentation strategy can generate more labeled samples, effectively supporting the training of the substitute model and reducing overfitting. As the number of labels increases, simple random augmentation can no longer satisfy the need for sample diversity, and the $\mathcal{L}_{\text{Grid}}$ strategy can solve this problem more effectively. For instance, at a 30k budget, the model performs best when $\omega = 0.4$. Further analysis indicates that regardless of the value set for ω, this

[1]https://cloud.tencent.com/product/tione
[2]https://www.kaggle.com/datasets/sumn2u/garbage-classification-v2

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TABLE VI
THE IMPACT OF THE OPTIMAL THRESHOLD VALUE ω

Queries	ω = 0.0	ω = 0.1	ω = 0.2	ω = 0.3	ω = 0.4	ω = 0.5	ω = 0.6	ω = 0.7	ω = 0.8	ω = 0.9	ω = 1.0
10K	69.44	69.75	70.10	70.75	71.00	71.42	72.24	**72.74**	72.60	71.82	70.64
20K	77.41	77.64	78.38	78.73	79.48	**79.72**	79.32	78.80	77.60	76.82	76.64
30K	81.24	81.58	81.73	82.03	**82.29**	81.90	81.68	81.34	81.17	80.84	80.76

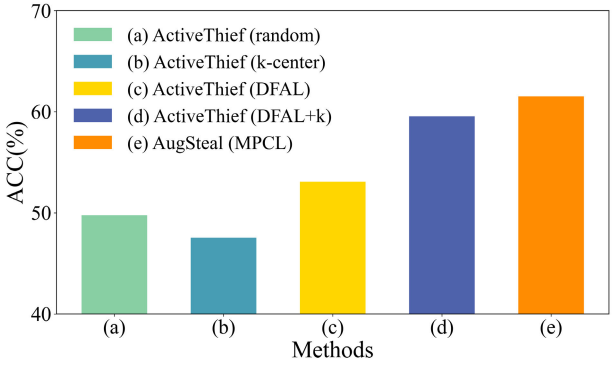


Fig. 13. The transferability of adversarial examples generated based on the substitute model.

method can lead to performance improvements. Moreover, this paper discusses alternative optimization strategies for the ω parameter to address the challenge of determining the optimal ω value under extreme conditions. When the query budget is limited, it is recommended to set ω to 0, relying entirely on $\mathcal{L}_{\text{Mul}}$ to guide training, which often yields superior results; conversely, when the query budget is ample, it is advisable to adjust ω to 1, allowing $\mathcal{L}_{\text{Grid}}$ to lead the training to foster better outcomes. This strategy provides a simple yet effective method for selecting the optimal ω value under extreme conditions.

F. Transferability of Adversarial Examples

Research has shown that adversarial samples generated against substitute models may also lead to misclassification by the original black-box model. Specifically, when implementing black-box adversarial attacks, a substitute model can be trained using the black-box model, and then adversarial samples can be generated based on this substitute model. These samples can often successfully transfer to the original black-box model, causing classification errors. In this study, we compare the AugSteal method with the ActiveThief algorithm and evaluate the transferability of adversarial samples generated by substitute models trained using AugSteal. To maintain consistency with the baseline paper ActiveThief, we conduct experiments on the MINST dataset with a query budget of 10k and use FGSM as the adversarial attack method with a perturbation rate of $\epsilon = 0.25$. We compare the transferability performance of adversarial samples generated using the AugSteal and ActiveThief methods. The experimental results are shown in Fig.13, where the method in parentheses indicates different active learning strategies. The results show that the AugSteal method improves transferability by 1.96 percentage points compared to the best results of ActiveThief, demonstrating the superior performance of AugSteal in generating highly transferable adversarial samples.

V. CONCLUSION

In this paper, we have comprehensively detailed the design philosophy, implementation specifics, and assessment outcomes of the AugSteal technique. Our exploration not only delves into the limitations of existing model steal technologies but also validates and analyzes AugSteal in the challenging realm of practical application scenarios. AugSteal comprises three innovative core components that collectively enhance the agreement of the substitute model in various aspects. Firstly, we developed a high-quality dataset optimization algorithm based on Grad-CAM, capable of selecting feature-rich and distinctive samples from extensive public datasets, thereby laying a robust foundation for model steal. Secondly, we incorporated the concept of active learning into the domain of model steal, innovatively proposing a sample selection mechanism that combines edge probability and data clustering. This mechanism is designed to select the most representative samples for different black-box models under limited query conditions. Lastly, we introduced a novel data augmentation technique, which, by integrating different loss function computation strategies, optimizes the training process, effectively addressing the issue of low agreement accuracy of the substitute model under restricted query numbers. This research successfully demonstrates that AugSteal can effectively extract black-box models even under multiple constraints, achieving satisfactory agreement accuracy. These results provide valuable insights for the future development of model steal strategies and aim to heighten model owners' awareness and defensive capabilities against security threats.

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